D03 · Industrial Automation & Robotics

01 · Industrial robots — welding, assembly, palletizing

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1 WHAT IT IS

Industrial robots are programmable mechanical arms applied to high-repetition, high-precision tasks — welding, assembly, palletizing, machine tending, painting, dispensing. The deployment requires three decisions in sequence: whether the case warrants robotization (volume, repeatability, geometry, payload), which robot architecture fits (6-axis articulated, SCARA, delta, gantry), and how the cell integrates with upstream/downstream operations. Most failed deployments inverted the order — bought the robot, then looked for the case.

2 WHY IT MATTERS

Industrial robots are the most visible automation investment a plant makes and the one most often deployed against the wrong case. **The dominant failure mode is robot deployment driven by capex availability or competitive imitation rather than by economic match — and a robot deployed against a case that does not fit becomes a $250K showpiece with $80K of annual maintenance and zero throughput gain**. The international safety standard that governs industrial robot deployment defines the engineering envelope **(ISO 10218-1:2011 + ISO 10218-2:2011 — international standards, "Robots and robotic devices — Safety requirements for industrial robots")**, and global deployment data from the International Federation of Robotics shows that **industrial robots achieve positive ROI within 24 months in roughly 75–85% of high-volume, low-mix cases, dropping to under 30% in low-volume, high-mix or variable-geometry cases**. The economics are unambiguous; the discipline is to identify which case is in front of you before committing the capex.

3 HOW IT IS DONE

1. **Apply the four-factor case-match test before considering any specific robot.** Volume (sustained throughput above the robot's economic floor, typically 50K+ cycles/year on a $250K robot), repeatability (the task as performed today must be definable in a deterministic program), geometry (workpiece variability within the robot's tolerance band), payload (within the manufacturer's spec with safety margin). A case that fails any of the four factors will not produce positive ROI no matter which robot is selected.
2. **Choose the robot architecture from the task, not from the vendor relationship.** 6-axis articulated robots (KUKA, FANUC, ABB, Yaskawa) for welding, assembly, complex motion; SCARA robots for high-speed planar pick-and-place; delta robots for very high-speed light-payload pick-and-place; gantry robots for large work envelopes or heavy payloads. Each architecture has a sweet spot; choosing across architectures based on vendor preference produces a cell that fights its own geometry.
3. **Engineer the end-of-arm tooling (EOAT) and the cell layout, not just the robot.** The robot is 30–40% of the cell cost; EOAT (grippers, weld torches, dispensers), fixturing, safety enclosures, conveyance, and integration are 60–70%. Plants that under-invest in EOAT and cell engineering produce robots that work in the lab and break in production. The EOAT and the workpiece-presentation strategy are where most cell failures originate.
4. **Design the safety system around ISO 10218 from the start — not as a retrofit after the cell is built.** Industrial robots are inherently dangerous (high speed, high payload, no force compliance); the safety system (fencing, light curtains, safety PLCs, emergency stops, lockout-tagout) must be integrated into the cell design, not bolted on after. Safety as retrofit costs 2–3x and produces cells that operators cannot service efficiently.
5. **Integrate the robot into the MES and the upstream/downstream flow — robots in isolation are throughput points, not throughput systems.** The robot produces parts; the cell delivers throughput; the integration with material supply, work order release, quality data capture, and downstream conveyance is what determines whether the throughput materializes. Plants that integrate the robot into MES capture data quality and traceability that justifies the capex; plants that operate the robot as a standalone PLC island produce parts but not visibility.
6. **Plan for the 6–12 month adoption curve — robot deployment is a process change, not an equipment purchase.** Operators must be retrained from manual-task workers to cell supervisors; maintenance must build robot-specific skills (most plants have no robot mechanics on day 1); engineering must build programming capability or contract it. The capex is paid in month 1; the operational value emerges over months 6–12 as the organization absorbs the new role structure. Plants that

under-budget the adoption curve produce robots that work but underperform.

WORKED EXAMPLE — ILLUSTRATIVE

An automotive Tier 2 supplier runs a manual MIG welding bay producing 4 brackets across 3 part families. Baseline: **47 brackets/hour throughput, 3.8% weld defect rate, $1.2M annual rework cost, 6 welders across 2 shifts**. The four-factor case-match test: volume = 89K cycles/year (above floor), repeatability = high (welds are deterministic), geometry = within ±0.4mm (within robot tolerance), payload = 3.2 kg (within spec). Case passes all four. Implementation: **2-robot cell** with FANUC ArcMate 100iC robots, custom EOAT, automated workpiece fixturing, and integration to MES for work-order release and quality data. Manual welding retained for 2 low-volume part families that fail the volume test. Results at 14 months: **throughput 47 -> 78/hour (+66%), defect rate 3.8% -> 0.4%, annual rework cost $1.2M -> $140K, robot cell payback = 18 months on a $740K total cell investment**. The 2 retained manual stations now handle the high-mix portfolio that the robot would have produced at 30% ROI; the architecture decision (robot for the 4 high-volume parts, manual for the 2 low-volume parts) is the move that converted the project from a marginal case into a clear win.

IN PRACTICE

Apply the four-factor case-match test before any vendor conversation

Volume, repeatability, geometry, payload — these four determine whether the case fits. IFR data shows industrial robots achieve positive ROI in 75–85% of high-volume / low-mix cases but under 30% in low-volume / high-mix or variable-geometry cases.

Engineer the cell, not just the robot

A robot does not produce parts; a cell does — EOAT, fixturing, conveyance, safety, integration. Plants that buy robots and try to build cells around them under-engineer the cell; plants that design cells and select robots for them produce the throughput the business case promised.

Design ISO 10218 safety from the start, not as retrofit

ISO 10218-1/-2:2011 defines the engineering envelope **(ISO 10218 — international standard, "Robots and robotic devices — Safety requirements")**. Safety retrofits cost 3–5× what designed-in safety costs and almost always degrade the cycle time the original case promised.

Operations leads case selection; engineering leads vendor selection

Capital planning drives timing only; vendor relationships drive nothing. Operations owns which case to robotize; manufacturing engineering owns architecture and integration; supply chain owns vendor negotiation. The plants that consistently produce successful deployments organize around this distinction.

WHAT YOU WOULD USE

INDUSTRIAL ROBOT PLATFORMS

FANUC / ABB / KUKA / Yaskawa (6-axis articulated) — The dominant industrial-robot OEMs with deep installed bases across welding, assembly, machine tending, and material handling; selection between them is typically driven by regional service support, integration partner availability, and incumbent automation footprint **(FANUC — vendor; ABB — vendor; KUKA — vendor; Yaskawa — vendor)**.

Epson / Mitsubishi / Stäubli (SCARA and high-speed pick-and-place) — Specialized architectures for high-speed planar tasks and precision assembly — strongest in electronics, pharmaceutical packaging, and small-parts assembly **(Epson — vendor; Stäubli — vendor)**.

Vendor-specific cell integrators (Comau, Genesis Systems, EWAB) — Pre-engineered cell integrators that pair a robot with EOAT, fixturing, safety, and integration as a turnkey solution; the right fit when in-house integration capability is limited **(Comau — vendor; Genesis Systems — vendor)**.

In-house engineering + robot OEM + qualified system integrator — For operations with manufacturing engineering capability — the OEM supplies the robot, a regional system integrator handles EOAT and safety, in-house engineering owns integration and program tuning. The path with the highest ceiling on long-term capability; requires the in-house engineering investment to be real.

4 SO WHAT

The question of which robot to buy is the answer to the wrong question — the right question is whether to robotize this case at all. Apply the four-factor case-match test before any vendor conversation, choose architecture from the task not from the vendor relationship, engineer the EOAT and the cell layout with the same rigor as the robot, design safety around ISO 10218 from the start not as retrofit, integrate the robot into the MES and the upstream/downstream flow, and budget the 6–12 month adoption curve as part of the project. The IFR success-rate data tells the whole story: the robots succeed where the cases match; the failures live in the cases that did not.

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